

- 1 Fig. 1.1 shows the structure of the amino acid glycine.

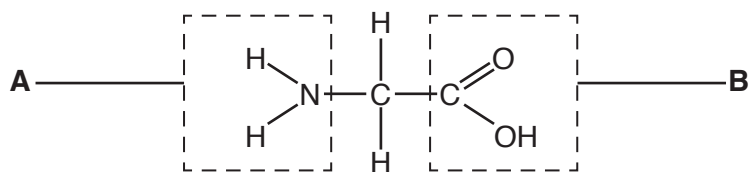


Fig. 1.1

- (a) (i) Name the parts of the amino acid molecule labelled **A** and **B** in Fig. 1.1.

A

B

[2]

- (ii) Amino acids are monomers used to build proteins.

Complete Fig. 1.2 by drawing a diagram to show the formation of a peptide bond between two molecules of glycine.

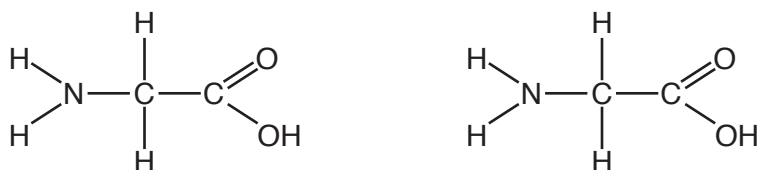


Fig. 1.2

[3]

- (b) Plasma cells synthesise and secrete antibodies.

Fig. 1.3 is a transmission electron micrograph showing a plasma cell.



Fig. 1.3

- (i) Use a label line and the label **T** on Fig. 1.3 to identify where the genes coding for the polypeptide chains of the antibodies are located. [1]
- (ii) Calculate the actual diameter of the plasma cell shown by the line **P–Q**.

Write down the formula used to make your calculation.

Show your working and give your answer to the nearest micrometre (μm).

formula

actual diameter = μm [2]

(iii) The plasma cell in Fig. 1.3 is very metabolically active.

Suggest why there are very few mitochondria visible in the electron micrograph in Fig. 1.3.

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..... [1]

(c) Sieve tube elements in plants have very few organelles such as mitochondria.

Explain how having very few organelles is an adaptation of the sieve tube element to its function.

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..... [2]

[Total: 11]